



Supercharge your work with Github Actions

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[@benmatheson](#)

ANCHORAGE
-team



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- *What is a Github Action*
- *How they work* 
- *Anatomy of an Action*
-   *Snow Tracker*
Demo
- *Starter Action*

A good way to
automatically run
code.

A good way to
automatically run
code.



in the cloud



A good way to
automatically run
code.



Github's servers



✨💧 “fresh,
newly-provisioned
virtual machine” ✨



**What can you
use Github
Actions for?**

What can you use Github Actions for?

1. Keep analysis current with regularly updating data 
2. Website builds 
3. Package development 
4. Ongoing testing, QA 



More inspiration

1. CRON Jobs
2. Webscraping
3. Rmarkdown reports
4. Social media bots
5. Email bots



PFDBOT

@pfdbot.bsky.social

Alaskans own 279,450 shares of Alphabet Inc worth \$43,658,474. It's in the 📺 Media & Entertainment industry, based in 🇺🇸 United States.

May 14, 2025 at 12:22 PM 👤 Everybody can reply



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Alaskans own 12,279 shares of Advanced Petrochemical Co worth \$100,986. It's in the 🚂 Materials industry, based in 🇸🇦 Saudi Arabia.

June 2, 2025 at 3:21 AM 👤 Everybody can reply





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Alaskans own 4,100 shares of Osaka Titanium Technologies Co worth \$52,751. It's in the 🏭 Materials industry, based in 🇯🇵 Japan.

May 13, 2025 at 6:23 AM 🗯️ Everybody can reply



PFDBOT

@pfdbot.bsky.social

Alaskans own 149,346 shares of China Resources Beer Holdings worth \$539,391. It's in the 🍕 Food Beverage & Tobacco industry, based in 🇭🇰 Hong Kong.

May 4, 2025 at 3:20 PM 🗯️ Everybody can reply





Alaska Highway Cameras 🏔️ 📡 📷
@alaskacameras.bsky.social

The view from Klawock Hwy @ Control Lake, 📷 from
511.alaska.gov



ALT



Alaska Highway Cameras 🏔️ 📡 📷
@alaskacameras.bsky.social

The view from Glenn Highway @ Gunsight Mountain MP 117, 📷
from 511.alaska.gov



ALT



Alaska Highway Cameras 🏔️ 📶 📷

@alaskacameras.bsky.social

The view from Seward Highway @ Bird Point MP 96.3, 📷 from 511.alaska.gov



ALT



Alaska Highway Cameras 🏔️ 📶 📷

@alaskacameras.bsky.social

The view from Tok Cutoff @ Mentasta Pass MP 79.2, 📷 from 511.alaska.gov



ALT

How actions work (*conceptually*)

Source: David Keyes: [R For the Rest of Us](#)

- Tell the code when it should run.
- Start up a virtual machine
- Install R
- Install all packages you need to run your code
- Run your code

Anatomy of an Action

snow_report_action.yml

Workflow 

Events 

Jobs 

Steps 

(do something)

Workflow



A YAML text document

Maps out when, what,
how of your action.

A giant to-do list.

Events



*What triggers the
code to run.*

Events



Pushing up code
Pull request
Issue created



Simple schedule
API call

Jobs



Wraps up steps.



All steps within job use
same runner - same virtual
computer.

Steps



Do the work.



Setup and run the code.

Anatomy of an Action

snow_report_action.yml

Workflow 

Events 

Jobs 

Steps 

(do something)

Demo:

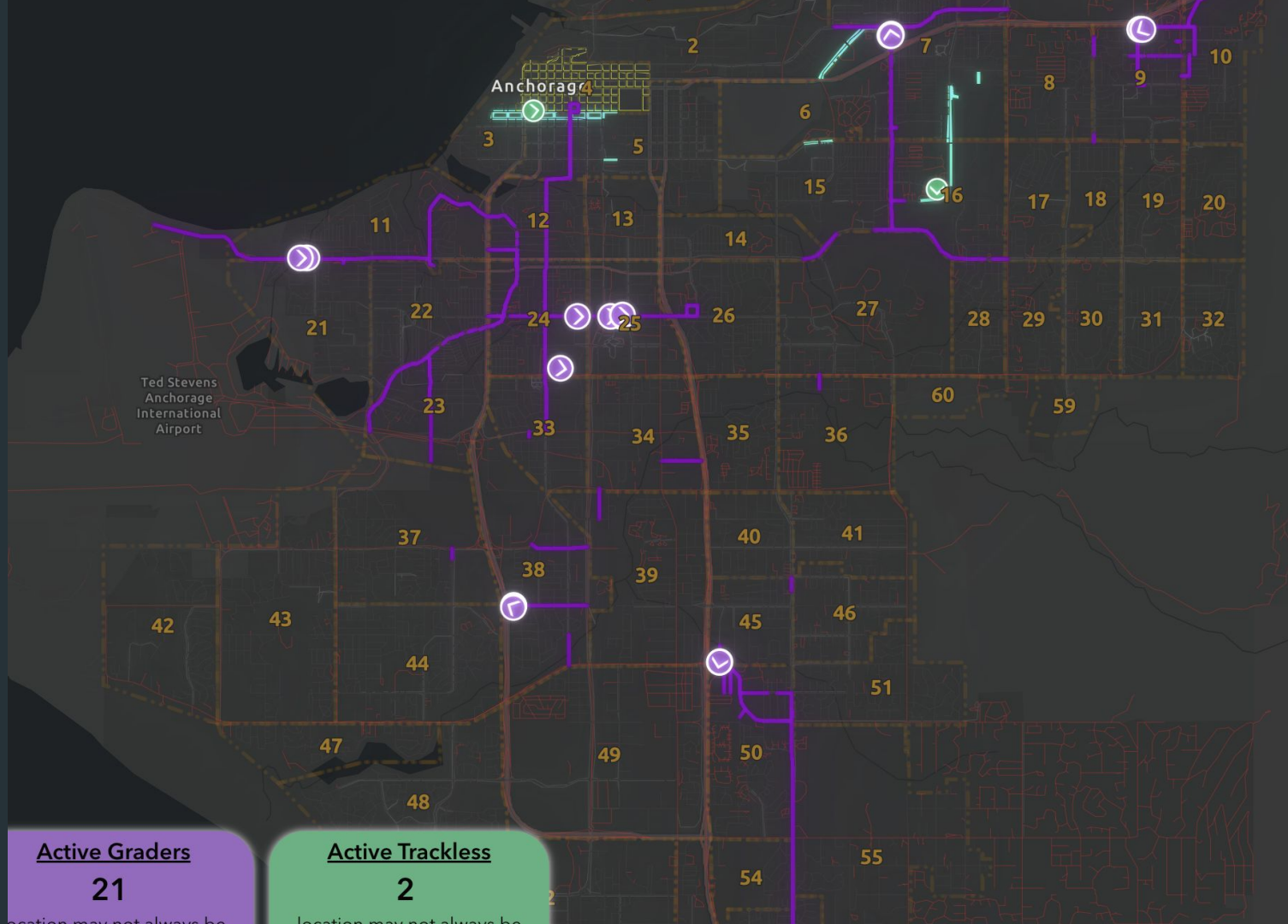
Anchorage Snowfall
tracking  

















(having
fun 🥵)



Anchorage Snowfall

PUBLISHED

June 17, 2025

Anchorage Ted Stevens International Airport Snowfall

 This report was run 2025-06-17.

 The source of the snowfall data is the [NOAA Nation Centers for Environmental Information](#) Climate Data Online (CDO) page. The site is shown here is Ted Stevens Anchorage International Airport.

2024-2025 in Anchorage cumulative snowfall compared to winters since 1952.

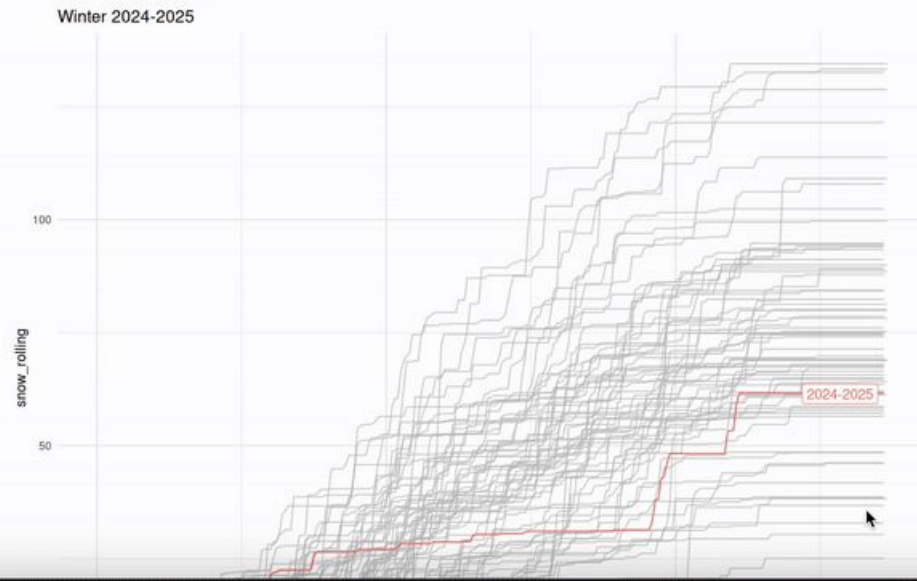


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[Largest Top 10 Years of Snowfall](#)

[Smallest Top 10 Years of Snowfall](#)

[Anchorage cumulative snowfall compared to winters since 1952 - Interactive Plot](#)

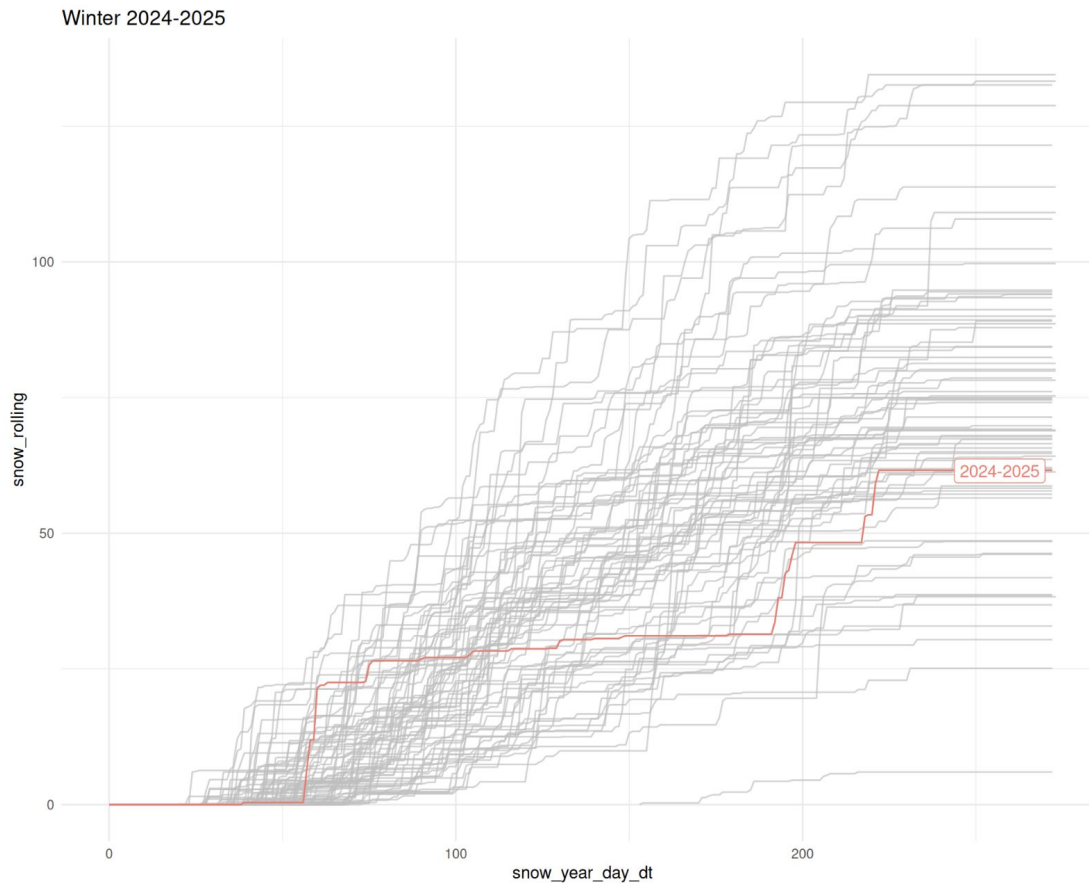
[Snow Day Patterns 2023-2024 and 2024-2025](#)

[2024-2025 Anchorage snow depth compared to winters since 1952.](#)

[2023-2024 Anchorage snow cumulative compared to the top 10 winters](#)

[Cumulative Snowfall-Decade Facets](#)

2024-2025 in Anchorage cumulative snowfall compared to winters since 1952.



Cumulative Snowfall- Decade Facets



Anatomy of an Action

snow_report_action.yml

Workflow 

Events 

Jobs 

Steps 

(do something)

snowfall_report

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

t

Add file

<> Code

actions-user snowfall ✓	36a29df · 17 hours ago	38 Commits
.github/workflows	Update snow_report_action.yml	last week
data/historic	initial	3 weeks ago
_quarto.yml	Add files via upload	3 weeks ago
description	Update description	3 weeks ago
snowfall.html	snowfall	17 hours ago
snowfall.qmd	prints	last week

README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)



benmatheson / snowfall_report

Q Type / to search



<> Code Issues Pull requests **Actions** Projects Wiki Security Insights Settings

Actions

New workflow

All workflows

pages-build-deployment

snowfall_report

Management

Caches

Deployments

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

Q Filter workflow runs



Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback



70 workflow runs

Event

Status

Branch

Actor



pages build and deployment

pages-build-deployment #31: by github-pages (bot)

main

17 hours ago

35s



snowfall_report

snowfall_report #39: Scheduled

main

17 hours ago

2m 49s



pages build and deployment

pages-build-deployment #30: by github-pages (bot)

main

yesterday

1m 52s



snowfall_report

snowfall_report #38: Manually run by benmatheson

main

yesterday

2m 26s



pages build and deployment

main

2 days ago





benmatheson / snowfall_report

Q Type / to search



<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

← snowfall_report

✓ snowfall_report #39

Re-run all jobs



Summary

Jobs

✓ create_report

Run details

Usage

Workflow file

Workflow file for this run



.github/workflows/snow_report_action.yml at af0e747

```
1
2 on:
3   schedule:
4     - cron: "0 0 * * *"
5   workflow_dispatch:
6
7 name: snowfall_report
8
9 jobs:
10
11 create_report:
12   runs-on: ubuntu-latest
13   env:
14     WXTOKEN: ${ secrets.WXTOKEN }
15
16   steps:
17     - uses: actions/checkout@v4
18
19     - uses: r-lib/actions/setup-r@v2
20     with:
21       r-version: '4.5.0'
```

Workflow



snow_report_action.yml



```
on:
  schedule:
    - cron: "0 0 * * *"
  workflow_dispatch:

name: snowfall_report

jobs:

  create_report:
    runs-on: ubuntu-latest
    env:
      WXTOKEN: ${ secrets.WXTOKEN }

    steps:
      - uses: actions/checkout@v4

      - uses: r-lib/actions/setup-r@v2
        with:
          r-version: '4.5.0'

      - uses: quarto-dev/quarto-actions/setup@v2
        with:
          version: 1.4.515

      - uses: r-lib/actions/setup-r-dependencies@v2
```

Events



```
name: snowfall_report
```

```
on:
```

```
  schedule:
```

```
    - cron: "0 0 * * *"
```

```
workflow_dispatch:
```

Jobs



```
jobs:  
  
  create_report:  
    runs-on: ubuntu-latest  
    env:  
      WXTOKEN: ${ secrets.WXTOKEN }
```


Steps



```
steps:
  - uses: actions/checkout@v4

  - uses: r-lib/actions/setup-r@v2
    with:
      r-version: '4.5.0'

  - uses: r-lib/actions/setup-r-dependencies@v2
    with:
      cache-version: 1

  - uses: quarto-dev/quarto-actions/setup@v2
    with:
      version: 1.4.515

  - uses: quarto-dev/quarto-actions/render@v2
    with:
      to: html
```

steps:

- uses: actions/checkout@v4

- uses: r-lib/actions/setup-r@v2
with:
r-version: '4.5.0'

- uses: r-lib/actions/setup-r-dependencies@v2
with:
cache-version: 1

- uses: quarto-dev/quarto-actions/setup@v2
with:
version: 1.4.515

- uses: quarto-dev/quarto-actions/render@v2
with:
to: html

steps:

- uses: actions/checkout@v4

- uses: r-lib/actions/setup-r@v2

with:

- r-version: '4.5.0'

- uses: r-lib/actions/setup-r-dependencies@v2

with:

- cache-version: 1

- uses: quarto-dev/quarto-actions/setup@v2

with:

- version: 1.4.515

- uses: quarto-dev/quarto-actions/render@v2

with:

- to: html

steps:

- uses: actions/checkout@v4
- uses: r-lib/actions/setup-r@v2
with:
r-version: '4.5.0'

- uses: r-lib/actions/setup-r-dependencies@v2
with:
cache-version: 1

- uses: quarto-dev/quarto-actions/setup@v2
with:
version: 1.4.515
- uses: quarto-dev/quarto-actions/render@v2
with:
to: html

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steps:
  - uses: actions/checkout@v4

  - uses: r-lib/actions/setup-r@v2
    with:
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  - uses: r-lib/actions/setup-r-dependencies@v2
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  - uses: quarto-dev/quarto-actions/setup@v2
    with:
      version: 1.4.515

  - uses: quarto-dev/quarto-actions/render@v2
    with:
      to: html
```

```
steps:
  - uses: actions/checkout@v4

  - uses: r-lib/actions/setup-r@v2
    with:
      r-version: '4.5.0'

  - uses: r-lib/actions/setup-r-dependencies@v2
    with:
      cache-version: 1

  - uses: quarto-dev/quarto-actions/setup@v2
    with:
      version: 1.4.515

  - uses: quarto-dev/quarto-actions/render@v2
    with:
      to: html
```


List of actions

1. [r-lib/actions/setup-r](#) - Sets up [R](#)
2. [r-lib/actions/setup-r-dependencies](#) - Installs packages declared in `DESCRIPTION`
3. [r-lib/actions/setup-renv](#) - Installs packages from renv lockfile
4. [r-lib/actions/setup-pandoc](#) - Sets up [pandoc](#)
5. [r-lib/actions/setup-tinytex](#) - Sets up LaTeX with [tinytex](#)
6. [r-lib/actions/check-r-package](#) - Runs `R CMD check` on an R package
7. [r-lib/actions/pr-fetch](#) - Fetches changes of a PR associated with an event
8. [r-lib/actions/pr-push](#) - Pushes changes to a PR associated with an event
9. [r-lib/actions/setup-manifest](#) - Sets up an R project with a Posit Connect `manifest.json` file



github.com/r-lib/actions

- name: Commit Data

run: |

```
git config --local user.email "actions@github.com"
```

```
git config --local user.name "GitHub Actions"
```

```
git add snowfall.html
```

```
git commit -m "snowfall" || echo "No changes to commit"
```

```
git push origin || echo "No changes to commit"-
```





Anchorage Snowfall

PUBLISHED

June 17, 2025

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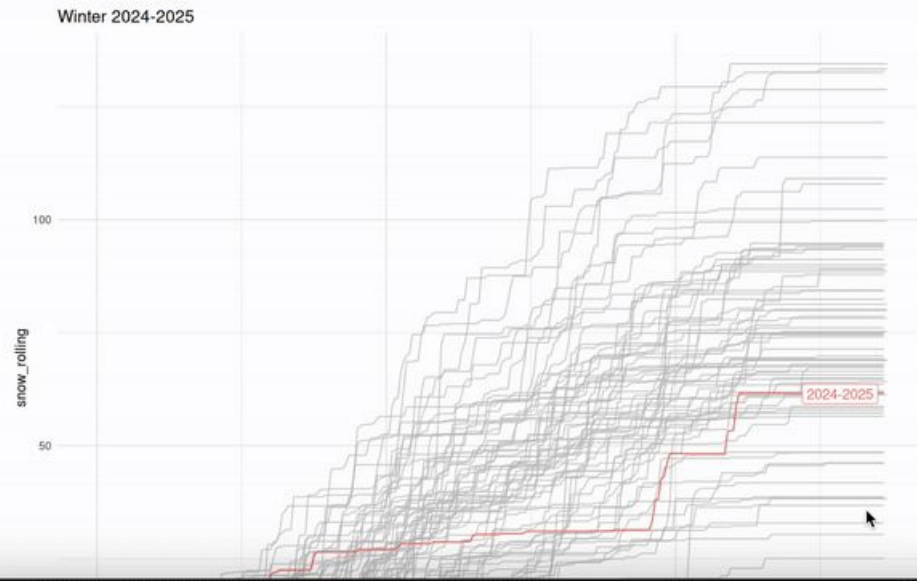


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[2024-2025 Anchorage snow depth compared to winters since 1952.](#)

[2023-2024 Anchorage snow cumulative compared to the top 10 winters](#)

[Cumulative Snowfall-Decade Facets](#)

Failure is
definitely an
option

☐	☆	▷	Ben Matheson 6	Inbox	[benmatheson/snowfall_report] Run failed: snowfall_report - main (...)	May 27
☐	☆	▷	Ben Matheson	Inbox	[benmatheson/snowfall_report] Run failed: snowfall_report - main (...)	9:57 PM
☐	☆	▷	Ben Matheson	Inbox	[benmatheson/snowfall_report] Run failed: snowfall_report - main (...)	9:32 PM
☐	☆	▷	Ben Matheson	Inbox	[benmatheson/snowfall_report] Run failed: snowfall_report - main (4684...	10:13 PM
☐	☆	▶	Ben Matheson 100	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (5...	May 27
☐	☆	▷	Ben Matheson	Inbox	[benmatheson/gh-pfdbot] Run failed: stock_tweet - main (a43829a) -	May 26
☐	☆	▷	Ben Matheson	Inbox	[benmatheson/snow-tracker] Run failed: render_snowfall - main (bcf0164)	May 24
☐	★	▷	Ben Matheson 2	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (1...	May 13
☐	☆	▷	Ben Matheson 100	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (5...	May 19
☐	☆	▷	Ben Matheson 21	Inbox	[benmatheson/gh-pfdbot] Run failed: stock_tweet - main (f74a8d4) - [be...	Apr 30
☐	☆	▷	Ben Matheson 70	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (bOc4a...	May 12
☐	☆	▶	Alaska Beacon	Inbox	AK Senate President Stevens to retire; Rep. Stutes to run for seat - Senat...	May 22
☐	☆	▶	Ben Matheson 9	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (b...	May 13
☐	☆	▷	Ben Matheson 4	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (3...	May 13
☐	☆	▷	Ben Matheson 3	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (1...	May 13
☐	☆	▷	Ben Matheson 100	Inbox	[benmatheson/alaskacameras] Run failed: cameras_tweet - main (b...	May 8

Job failure rate

100%

Failure rate across jobs in this organization for current month

Job failure rate

1



Failure rate across jobs in this organization for current month



Failure rate

Failure rate across jobs in this organization for current month

Actions

New workflow

All workflows

pages-build-deployment

snowfall_report

Management

Caches

Deployments ↗

Attestations ↗

Runners

Usage metrics ↗

Performance metrics ↗

All workflows

Showing runs from all workflows

Q Filter workflow runs

Help us improve GitHub Actions

Tell us how to make GitHub Actions work better for you with three quick questions.

Give feedback

×

70 workflow runs

Event ▼ Status ▼ Branch ▼ Actor ▼

✓ pages build and deployment pages-build-deployment #31: by github-pages bot	main	📅 17 hours ago 🕒 35s	...
✓ snowfall_report snowfall_report #39: Scheduled	main	📅 17 hours ago 🕒 2m 49s	...
✓ pages build and deployment pages-build-deployment #30: by github-pages bot	main	📅 yesterday 🕒 1m 52s	...
✓ snowfall_report snowfall_report #38: Manually run by benmatheson	main	📅 yesterday 🕒 2m 26s	...
✓ pages build and deployment	main	📅 2 days ago	...



benmatheson / snowfall_report

Q Type / to search



<> Code Issues Pull requests **Actions** Projects Wiki Security Insights Settings

← snowfall_report

✓ snowfall_report #39

Re-run all jobs



Summary

Jobs

✓ create_report

Run details

Usage

Workflow file

Triggered via schedule 17 hours ago

Status

Total duration

Artifacts

benmatheson af0e747 main

Success

2m 49s

—

snow_report_action.yml

on: schedule

✓ create_report

2m 45s





benmatheson / snowfall_report

Q Type / to search



<> Code Issues Pull requests **Actions** Projects Wiki Security Insights Settings

← snowfall_report

✓ **snowfall_report #39**

Re-run all jobs



Summary

Jobs

✓ **create_report**

Run details

Usage

Workflow file

create_report

succeeded 17 hours ago in 2m 45s

Q Search logs



> ✓ Set up job	3s
> ✓ Run actions/checkout@v4	1s
> ✓ Run r-lib/actions/setup-r@v2	1m 44s
> ✓ Run quarto-dev/quarto-actions/setup@v2	8s
> ✓ Run r-lib/actions/setup-r-dependencies@v2	28s
> ✓ Run quarto-dev/quarto-actions/render@v2	14s
> ✓ Commit Data	3s
> ✓ Post Run r-lib/actions/setup-r-dependencies@v2	0s
> ✓ Post Run actions/checkout@v4	0s
> ✓ Complete job	0s

Summary

Jobs

create_report

Run details

Usage

Workflow file

create_report

succeeded 17 hours ago in 2m 45s

Search logs



✓	Run quarto-dev/quarto-actions/setup@v2	8s
2442	(Reading database ... 85%	
2443	(Reading database ... 90%	
2444	(Reading database ... 95%	
2445	(Reading database ... 100%	
2446	(Reading database ... 229027 files and directories currently installed.)	
2447	Preparing to unpack .../quarto-1.4.515-linux-amd64.deb ...	
2448	Unpacking quarto (1.4.515) ...	
2449	Setting up quarto (1.4.515) ...	
2450		
2451	Running kernel seems to be up-to-date.	
2452		
2453	No services need to be restarted.	
2454		
2455	No containers need to be restarted.	
2456		
2457	No user sessions are running outdated binaries.	
2458		
2459	No VM guests are running outdated hypervisor (qemu) binaries on this host.	
2460	Quarto Installed !	
>	✓ Run r-lib/actions/setup-r-dependencies@v2	28s
>	✓ Run quarto-dev/quarto-actions/render@v2	14s
>	✓ Commit Data	3s
>	✓ Post Run r-lib/actions/setup-r-dependencies@v2	0s
>	✓ Post Run actions/checkout@v4	0s
>	✓ Complete job	0s

► Run # Install quarto

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Reading package lists...

Building dependency tree...

Reading state information...

The following packages will be DOWNGRADED:

quarto

0 upgraded, 0 newly installed, 1 downgraded, 0 to remove and 31 not upgraded.

E: Packages were downgraded and -y was used without --allow-downgrades.

Error: Process completed with exit code 100.

✓ Commit Data

```
1 ▶ Run git config --local user.email "actions@github.com"
16 [main 2f08f56] snowfall
17 1 file changed, 5979 insertions(+)
18 create mode 100644 snowfall.html
19 remote: Permission to benmatheson/snowfall_report.git denied to github-actions[bot].
20 fatal: unable to access 'https://github.com/benmatheson/snowfall_report/': The requested UR
21 No changes to commit-
```

> ✓ Post Run r-lib/actions/setup-r-dependencies@v2



TOP SECRET

Repository secrets

[New repository secret](#)

Name 	Last updated
 WXTOKEN	6 minutes ago  



TOP SECRET

Actions secrets / New secret

Name *

Secret *

Add secret



TOP SECRET

```
wxToken <- Sys.getenv("WXTOKEN")
```

R Github Actions Starter



github.com/benmatheson/r-github-actions-starter

benmatheson / r-github-actions-starter

Q Type to search

<> Code

Issues

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benmatheson

r-github-actions-starter

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Fork 0

Star 0

main 1 Branch 0 Tags

Go to file

Add file

<> Code

About

benmatheson

Update README.md

3eaccb4 · last week

6 Commits

.github/workflows	Update starter_workflow.yml	2 weeks ago
README.md	Update README.md	last week
description	Add files via upload	2 weeks ago
my_script.R	Add files via upload	2 weeks ago

README



This repo includes a basic starter template for running a Github action with R.

This will install R, a list of packages, and provide options for installing and rendering quarto, as well as comitting data at the end of the workflow.

a few important files that are used here:

`description` - lists the packages to install.

`my_script.R` - the R script that will be run.

In the `.github/workflows/starter_workflow.yml` file, the following steps are taken.

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Languages

R 100.0%

```
# This template is a starter workflow to get the basics of R, R
packages, and optional quarto installed
```

```
name: Starter R workflow
```

```
# Controls when the workflow will run - using a cronjob in this
case
```

```
on:
```

```
  # Triggers the workflow on a schedule - this runs at one
  minute past each hour.
```

```
  schedule:
```

```
    - cron: "1, * * * *"
```

```
  workflow_dispatch:
```

```
# A workflow run is made up of one or more jobs that can run
sequentially or in parallel
```

```
jobs:
```

```
  run-r-script:
```

```
    runs-on: ubuntu-latest
```

```
    env:
```

```
      VERY_SECRET_KEY: ${ secrets.VERY_SECRET_KEY }}
```

```
    ##create a secret if you need to use it in the code
```

```
  # Steps represent a sequence of tasks that will be executed
  as part of the job
```

```
  steps:
```

```
    # Checks-out your repository under $GITHUB_WORKSPACE, so
    your job can access it - makes files accessible to this
    workflow
```

```
    - uses: actions/checkout@v4
```

```
  #install R
```

```
    - uses: r-lib/actions/setup-r@v2
```

```
      with:
```

```
        r-version: '4.5.0' #can specify the R version you
```

```
want
```

```
  #install dependencies in the "description" file
```

```
    - uses: r-lib/actions/setup-r-dependencies@v2
```

```
      with:
```

```
        cache-version: 1
```

```
  # option to install quarto and a quarto render step
```

```
  # - uses: quarto-dev/quarto-actions/setup@v2
```

```
  #   with:
```

```
  #     version: 1.4.515 #set version here
```

- install packages
- run r script
- save data
- (quarto)

Favorite Resources

R For The Rest of Us Tutorial:

<https://rfortherestofus.com/2023/05/github-actions>

Bea Milz Tutorial:

<https://beamilz.com/posts/series-gha/2022-series-gha-1-what-is/en/>

R Actions to use

<https://github.com/r-lib/actions>



Supercharge your work with Github Actions

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